

National University of Computer and Emerging Sciences, Lahore Campus



Course: Digital Logic Design Lab Course Code: EL-1005
Program: BS(EE) Semester: Spring 2024 Duration: 2 Weeks Total
Marks: 50

Section: All Weight 30%

CLO CLO-4

PROJECT STATEMENT:

Implement the project assigned to you on both software (Logic Works) and hardware (on breadboard) in groups of 2 students.

Digital Clock

A digital clock is an alternative to a traditional analogue clock. This type of clock shows numbers to display the time in a digital format, such as on a watch, phone, or an alarm clock. This can be in both 12 and 24-hour formats. You have to design the digital clock using the 7-segment display that can also perform the following function.

- If SET *minutes* is pressed a minute should be updated to set the desire minutes of time.
- If SET *hours* is pressed a minute should be updated to set the desire hours of time.

All the operations mentioned above should be performed using the combinational and sequential circuits. You can use the BCD to seven segment converters if required. You can either use a 12- or 24-hours format. If you are using the 12 hours format than a separate led should be used that will work as an indicator of the AM/PM.

Project Deliverables:

Prepare a detailed report on the completion of project which will summarize all the relative information related to different tasks of the project.

- Report Submission on Google Classroom.
- Hardware Evaluation in your respective lab sessions. (Bring hardcopy of report during evaluation)

Deadline: 29th April-3rd May 2023